Next Gen Email app with advanced voice recognition for visually impaired

Josephinleela R
Professor
Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
pecleela2005@gmail.com

Jagan G
UG Scholar
Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
jaganchandru1568@gmail.com

Jagadeesh K
UG Scholar
Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
jagadeeshkugan2702@gmail.com

Chiranjeevi R
UG Scholar
Department of Computer Science and Engineering
Panimalar Engineering College
Chennai, India
rchiranjeevichiru459@gmail.com

***Abstract*- The internet is a common way for all people to acquire knowledge and information. Communication is one of the key areas where the Internet has altered society. However, using any email-based service and accessing these text documents might be challenging for blind individuals. Blind persons have benefited greatly from the audio feedback based virtual environments, such as screen readers, that enable them to use online apps. We outline the design of the voicemail system, which enables a blind person to effectively and conveniently retrieve emails. With the aid of this initiative, blind users can now use voice commands to access any documents or other resources in our system. Therefore, in an effort to improve society and grant these individuals with disabilities equal status, we have developed this project idea that enables users to transfer emails using voice instructions in place of a keyboard or other visual aids.**

***Keywords*- E-mail; voice mail; Email for blind; Visually impaired**

I. INTRODUCTION

As we all know, the advent of the Internet has transformed a number of fields. People's lives have become so much easier thanks to the Internet that they may readily get whatever information they want. One of the primary areas that the Internet has significantly altered is communication. When it comes to delivering and receiving crucial information via the Internet, emails are the most reliable mode of communication. However, in order for humans to use the Internet, they must meet a particular standard, which requires vision. However, there are also those in our society who are not as fortunate as you. Some people are blind or visually handicapped, meaning they are unable to see objects, including the keyboard and computer screen. Over 240 million individuals worldwide are visually impaired, according to a poll. In other words, almost 240 million people do not know how to utilize email or the Internet. The sole method for a visually impaired person to send an email is for them to speak the full message to a non-visually impaired person, who will then write and send the email on the visually impaired person's behalf. However, this is not the best approach to take on the issue. It is highly improbable that an individual with vision impairments will always be able to locate assistance. Despite the fact that our culture criticizes visually handicapped persons for these reasons. Thus, we have developed this project concept that enables the user to send emails using voice instructions without requiring a keyboard or any other visual aids. Voice command interactions can underpin any action. Furthermore, since the gadget will recognize and process voice instructions, the blind user need not worry about knowing which mouse clicks to do to utilize a certain service.

II. LITERATURE SURVEY

[1] **Rajini S, Dhivyaa A S, Jananipriya R M, Karthika priya V S**, proposed a paper in which voice-based email for the visually impaired people was created as an application to benefit the visually impaired by allowing them to share and communicate messages without typing. They created the application in such a way that they use the speech to text and text to speech translators, allowing users to control their email accounts through their voice. Their system lets the users use it with voice commands to do certain actions and the user will reply to the same. Speech to Text, also called Automatic Speech Recognition, converts speech into text, allowing you to compose emails. IVR methodology is used to allow computers to communicate with human beings through voice and DTMF tones through the keyboard.

"VOICE-BASED EMAIL SYSTEM FOR VISUALLY IMPAIRED PEOPLE June-2022 "

Advantages-It reduces the use of keyboard shortcuts and mouse keys which are difficult to handle and remember for the blind people. This reduces the traditional way of sending mail as recordings for blind. Commands such as ‘Inbox’, ‘Sent’, ‘Compose’, and ‘User information’ were used and their corresponding actions will be performed. This reduces the burden on the users.

Disadvantages-Videos, Audios, Image files can't be embedded in this system. Also, there is no encryption and decryption algorithm used here. So, sending any credentials via this system will be risky for the users.

[2] **Neha Gupta, Nitin Chauhan, Nirbhay Pal, Mayank Bhatnagar, Mohd Fardeen** released a paper for voice-based email to help blind people to access and write emails in a problem-free manner. This system greatly reduces reliance of visually impaired people on others for their activities regarding emails. They are proposing smartphone-based applications working on either IOS or android operating systems. They use some modules to make voice-based email such as Event Listener, Command Module, Authenticator etc... When user opens app for the first time he/she is prompted to ask to their family member or closed one to help them is setup process as host Oss requires physical intervention of user to grant permissions required by the application to work properly on host OS, apart from that user id and password needs to be inputted manually and biometric authentication setup is also done here. They also use voice commands like create emails, read email, change folder, search etc....

“DRISHTI: VOICE BASED EMAIL SYSTEM FOR VISUALLY IMPAIRED July-2021”

Advantages- This system is designed to bridge the social gap between visually impaired individuals and those without visual impairments, enhancing the lifestyle and employment opportunities for the unsighted. Accessible to all age groups, the application is user-friendly for everyone, including the visually compromised, the illiterate, and people with normal vision, thereby promoting inclusivity and accessibility for a diverse user base.

Disadvantages-Multimedia files can't be embedded in this system. There is no cryptographic algorithm used here. So, sending any credentials or personal via this system will be risky for the users.

[3] **Dr. K Badari Nath, Ranjith kumar M E, Nethravathi T, Aman Kumar** proposed a Blind Friendly-Email-System that enables visually impaired and illiterate people to use day-to-day technologies like sending and receiving emails. Their system supports E-Mail Compose, Inbox Accessing and reading/listening recieved mails. They also provided a login system that works on the principle of speech recognition and touch enabled standalone system, which was built using blind people friendly GUI forms. Additionally, they implemented imaplib, smtlib, Google's STT and PyAudio for protocols to send and receive emails and to implement audio supports.

"Voice Based Email System For Visual Impaired Using AI June 2023 "

Advantages-They provided multi language support that enables users across multiple regions and countries to use the system. Trained with multiple audio recordings and included acoustic modelling.

Disadvantages-This proposed system is only suitable for desktop applications and no support for vast number of users who were mobile users.

[4] **Gagana M, Brundha K, Nita meshram** proposed a system that makes use of IVR technology and Python libraries. Blind persons have benefited greatly from the use of audio feedback based virtual environments such as screen readers in accessing online apps. In order to relieve the user of the burden of remembering which mouse click action they wish to do, this system offers the location of the user's voice prompts. They use both Speech To Text and Text To Speech methodology for the system works by users' voice commands. And also, it includes IVR technology for other purposes. Here they use voice notification of buttons like if mouse arrow place or across the any button, it will give voice notification to the user.

"VOICE BASED EMAIL FOR VISUALLY IMPAIRED-2023"

Advantages-To provide voice-based mailing service where they could read and send mail on their own. It provides a low-cost system that will be able to automatically locate and read the text. allowed visually impaired persons. It designs and analyzes the pre-processing modules for text recognition. It designs and analyzes the segmentation process for extracting and resizing letters.

Disadvantages-This proposed system is only suitable for desktop applications and has no support for the vast number of users who were mobile users. It uses mouse so, it also a difficulty to the visually impaired people.

[5] **Amritha Suresh, Binny Paulose, Reshma Jagan, Joby George** created a voice-based email system that will be easy in use for visually impaired. Their work aims to reduce the software load of using screen readers and automatic speech listener and the user’s cognitive load of remembering keyboard shortcuts. They made a login system in which the login credentials were entered by means of speech. Additionally, the commands were given as instructions for the users to utilize this system.

"Voice Based Email for Blind - 2016"

Advantages-This system provides voice-based register and login. Additional options 0to view sent messages and trash, message retrieval was also included in this project.

Disadvantages-There is no secure login system, and no encryption methods were involved. The system works on the desktop only and not on mobile phones. This system is implemented only on localhost as a web application.

[6] **Nivedita Bhore, Shraddha Mahala, Komal Acharekar, Dr. Madhavi Waghmare** proposed the desktop application of voice-based email. In this system, certain keyboard shortcuts will be employed. Other than that, there is no need to utilize a keyboard. The developer will have already saved the user's credentials in the system. Therefore, the user won't have to repeatedly log into the program. Depending on the activity, the speech engine will continuously ask the user to undertake the operation.

"Email System for Visually Impaired People -2021"

Advantages-One of the important advantages of this system is that the user need not use the mouse since all the operations will be mostly based on voice commands and certain keyboard shortcuts at some places. This entire system is based on speech-to-text and text-to-speech majorly. When the user runs the application, a welcome page will be displayed.

Disadvantages-Still, it uses keyboard shortcut, and it is desktop application. In this application, we have no way to add attachment of documents, images, audios etc., through voice commands.

[7] **Paulus A.Tiwari , Pratiksha Zodwan , Harsha P.Nimkar, Trishna Rotke, Priya G.Wanjari, Umesh Saarth** Introduces voice-activated email software, such as interactive voice response (IVR), which allows users to communicate with PCs via voice, and automated speech recognition (ASR), sometimes referred to as speech-to-text converter. technology that enables users to interact with computers using their voices in a manner similar to natural human speech. Blind individuals are encouraged to use voice messaging engineering to access emails and other interactive media capabilities, in addition to other functional framework functions. Here, the author demonstrates how to create a voice message that blind users may use to access emails and other material inside a productive and successful framework.

"A Review on voice based email system for blind - June 2020"

Advantages-Multimedia access and attachments were also provided with this software. This lets users share media files with just a few voice commands.

Disadvantages-There was no implementation of security algorithms in message transfer. Also, this project had reduced the keyboard and mouse usage only to an extent. They focused only on the desktop users.

[8] **Sanjeevi Kumar P, Ashwitha Shetty, Megha Manjunath Naik, Nayak Ashmitha Suresh, Sachin** proposed the voice based working email system. In the System, document selection features were also available. If the user forgets the password, in which the authorized person forgets the password and cannot log in all the time, the user can choose the forgotten password module. In this module, the user may first be asked to enter a username. The security question can be searched on the home page by user ID. This is a registration question. The website has unlimited development possibilities. Instead of verifying the user with an email address and password, enhancements such as a fingerprint reader can be used. Additional voice commands can be added to improve the visual interface, such as answer, benefit, join, star, etc..

"VOICE BASED EMAIL SYSTEM FOR VISUALLY IMPAIRED MAY 2022"

Advantages-Here, they use bio-metric authentication. So, users can easily handle this system. This system also verifies the receiver side user for security purposes. This can help a person with the right information and get the email they want.

Disadvantages-There is no secure way of encryption methods were involved. The system works on the desktop so blind people have some difficulties to use system.

[9] **Onkar Indalkar, Sampada Ghorpade, Atharva Sonone, Kishor Pathak** in their review, suggested a project which is based on voice email that will help impaired people to communicate just as normal people. Authors proposed this system with one idea in mind that it has to be easily accessible for all kinds of persons. Thus, in this system the PC is going to be promoted to let the user perform specific operations. Then the user must register using the registration form and the user will be assisted through voice commands. Registered users can login by speaking password and username when prompted by system. They said that the login credentials are stored as voices in the database, thus there is no conversion of speech to text for logging in.

"REVIEW ON VOICE BASED EMAIL SYSTEM FOR BLIND PEOPLE - Mar 2022"

Advantages-Their system can be easily applied in the problem of Text/instance retrieval. They allow users to hear the recently received mails to the Inbox, as well as the IVR (Interactive Voice Response), technology proves very effective for them in terms of guidance.

Disadvantages-Their system cannot detect features from low quality Text. Also, poor detection or false detection occurs from low quality voice.

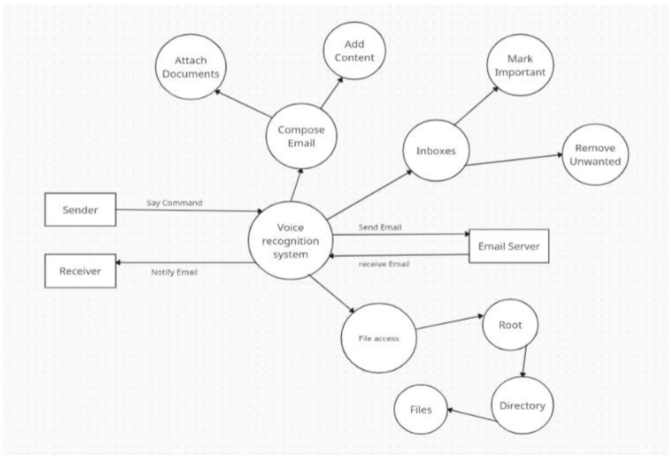

fig no. 1 Data Flow Diagram

III. EXISTING SYSTEM

The discussion revolves around the accessibility and functionality of email systems, particularly for visually impaired users. Initially, it notes that basic email frameworks have been designed with accessibility features that rely heavily on voice commands and text-to-speech systems. These are intended for users to navigate and control their email functions without the traditional use of a keyboard, by memorizing keyboard shortcuts.

However, the critique of the current state of voice-based email systems is highlighted. These systems often employ Interactive Voice Response (IVR), speech-to-text converters, mouse click events, and screen readers to assist users. Despite these features, the systems fall short in several aspects,

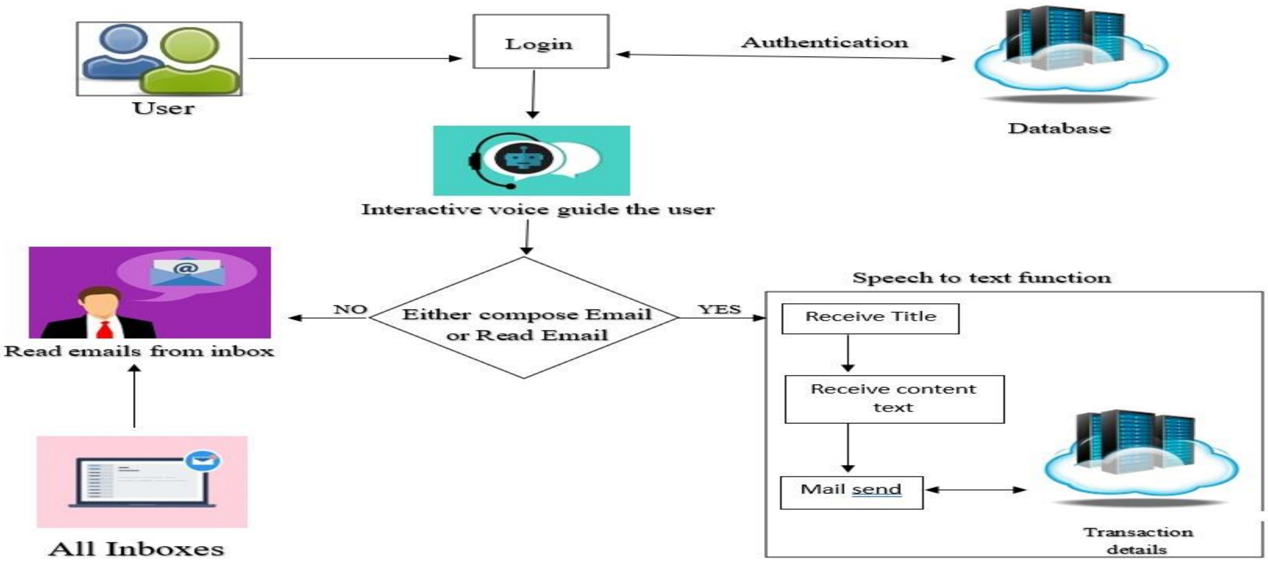

fig no. 2 System Architecture

especially when evaluating their utility for visually impaired individuals. A significant issue is the method of sending emails and their content as audio files, which proves to be impractical for those who rely on audio feedback for communication.

The core of the concern is that these email systems do not adequately support blind users. There is a lack of audio feedback, which is critical for visually impaired users to navigate and understand the content and functions of the email system effectively. Such a shortfall means that performing fundamental email tasks like composing new emails, checking the inbox, managing the trash folder, and viewing sent items becomes a challenge, if not impossible, for visually impaired users.

Moreover, some email applications attempt to incorporate accessibility by including a microphone icon for voice commands. By tapping this icon, users can speak their messages, which the system then converts into text format. While this feature aims to enhance accessibility, it still does not fully cater to the needs of visually impaired individuals. These users require more comprehensive audio feedback and navigation options to interact with email systems efficiently. The inability to 'see' and 'read' in the conventional sense means that visually impaired users depend heavily on audio cues and feedback, which are lacking in the systems described.

In essence, the current landscape of voice-based email systems demonstrates a significant gap in accessibility and usability for visually impaired users. Despite some attempts to make email communication more accessible through voice commands and speech-to-text technology, these systems are yet to provide a fully inclusive and functional experience for all users, particularly those who are blind or have low vision. The need for innovative solutions that offer comprehensive audio feedback and intuitive navigation capabilities is critical in making email communication truly accessible to everyone.

Drawbacks: Client need to utilize mouse associated with the PC and ought to perform mouse click occasions to send and get messages.

IV. Proposed System

The proposed system heralds a transformative step forward in communication technology, focusing on a highly innovative feature: voice command functionality. This paradigm shift away from conventional input methods, such as mouse clicks and keyboard strokes, towards an intuitive voice command interface, marks a significant departure from the norm. It offers an unparalleled ease of use, allowing users to execute operations through simple voice commands. This eliminates the cumbersome necessity of memorizing keyboard shortcuts or navigating through complex menus, making the technology not just user-friendly, but also universally accessible.

A critical aspect of this system is its dedication to inclusivity. By prioritizing voice commands as the primary mode of interaction, our system opens up a world of possibilities for individuals with disabilities, ensuring that they, too, can benefit from digital advancements without facing traditional barriers. This inclusivity extends the system's utility across a diverse user base, promising an enhanced user experience for everyone, regardless of their physical abilities.

In the realm of functionality, our system excels in integrating voice command technology across all facets of information technologies and communication tasks. Users can effortlessly perform a wide array of actions, such as sending emails, browsing their inboxes, and attaching documents, all through spoken commands. This hands-free operation is not only a boon for efficiency but also substantially minimizes the likelihood of manual errors that are common with traditional data entry methods.

Moreover, the system's versatility across different platforms, including both mobile devices and computers, ensures that users have the flexibility to access and manage their email from anywhere, at any time. This cross-platform compatibility further enhances the user experience by providing seamless interaction with the system, regardless of the device in use.

In essence, our proposed system stands as a beacon of innovation in communication technology, offering a brilliantly user-friendly, inclusive, and efficient solution. It redefines the way users interact with technology, making it more accessible and practical for a broad spectrum of users and setting a new standard for what communication systems can achieve. This system is not merely an advancement but a significant leap toward making technology universally accessible and efficient in a way that has never been seen before.

V. METHODOLOGY

a) Speech To Text

Speech recognition mainly focuses on converting spoken language into written text or computer commands. Speech recognition is commonly used for tasks such as transcription services, voice assistants, etc. Nowadays, speech recognition uses hidden Markov models. The analog voice signal must first be sampled along the time and amplitude axes or digitized. The voice signal samples are analyzed at equal time intervals. This interval is typically 20 ms because the signal during this interval is considered static. Speech-to- text translation is performed using google's voice recognition feature, which requires an active internet connection. Google Speech to Text is one of the easy and effective ways to listen to user commands and convert them into text accurately. In our system, the user's voice commands are heard directly from the microphone. The user's voice commands are collected using a listen command that takes a source parameter set by our system's microphone.The voice is converted to text using the recognize_google command to convert the audio file to text. These functions are available in the Recognition function of Python's Speech_recognition package.

b) Text To Speech

Text-to-Speech is also known as Speech Synthesis. For this we use pyttsx python package to implement in our project. pyttsx is a cross-platform, platform-independent text-to-speech library. The main advantage of using this library for text-to-speech conversion is that it works offline. This makes it more reliable to use for voice-based projects. We can choose among different voices that are installed on the system, Controlling the speed of speech by this package. Although it was originally used by blind people to listen to written documents, it is now widely used to transmit financial data, electronic messages, and other information over the phone to people.

c) SMTP

Simple Mail Transfer Protocol (SMTP) is used to send and receive emails. It is sometimes combined with IMAP or POP3 to handle message retrieval, while SMTP primarily sends messages to the server for transmission. SMTP is an asymmetric protocol, meaning multiple clients interact with a single server, using a basic model popular in the 1980s, now largely defunct beyond email protocols. SMTP works over TCP/IP.

d) IMAP

Internet Message Access Protocol (IMAP) is the standard "incoming" email retrieval protocol. It stores emails on the mail server and allows recipients to view and manipulate them as if they were stored locally on their device. IMAP acts as an intermediary between the mail server and the mail client. When users read email via IMAP, they read it on the server. They don't actually download or store emails on their local device. This means that email is not tied to a specific device and users can access it from any location in the world using different devices.

e) POP

Post Office Protocol (POP3) provides access to the inbox stored in the email server. It performs message download and deletion operations. So when a POP3 client connects to a mail server, it retrieves all mail from the mailbox. It then stores them on your local computer and deletes them from the remote server. Modern POP3 clients allow you to keep a copy of your messages on the server if you explicitly select this option. There is no limit to the size of emails we receive or send. This requires less storage space on the server because all messages are stored on the local machine. Transferring mail folders from a local machine to another machine can be difficult.

VI. CONCLUSION

Our application is created by our team remembering the challenges of the visually debilitated, assisting them with utilizing email administrations. It is our responsibility to provide them a medium which provides the comfort of exchanging emails with ease and security. Screen readers have been eliminated. This makes the framework significantly more viable, actually tested people will actually want to connect with the rest of the world. Despite age, everybody can utilize the proposed email framework effortlessly. It has speech to text elements as well as text to speech highlights. With this, individuals who are visually impaired or outwardly impeded can send and get mail effectively with simply voice orders and extremely negligible mouse utilization.

VII. REFERENCE

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